WGAN-based Data Reconstruction Attack (GLASS)

Background

- Split inference enables edge-cloud collaboration by leaking intermediate representations (IR).
- GLASS reconstructs private data from IR using generative models.
- We evaluate WGAN for improving reconstruction quality.
- DISCO is used to assess model performance with defense enabled.

Objectives

- Use WGAN to replace StyleGAN in GLASS.
- Integrate DISCO defense to evaluate models performance.
- Evaluate performance using PSNR and LPIPS.

Dataset & Setup

- Dataset: CIFAR-10
- Framework: PyTorch
- Metrics: PSNR, LPIPS

Loss Functions

$$\mathcal{L}_{recon} = \|x - \hat{x}\|_2^2 \quad (1)$$

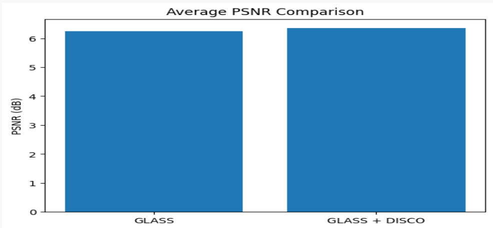
$$\mathcal{L}_{WGAN} = \mathbb{E}[D(x)] - \mathbb{E}[D(\hat{x})] \quad (2)$$

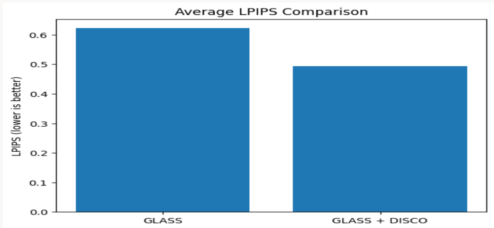
- Total Loss: $\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{recon} + \lambda_2 \mathcal{L}_{WGAN}$

Visual Results

- DISCO improves image sharpness and structure.
- Visual similarity to original increases with DISCO.

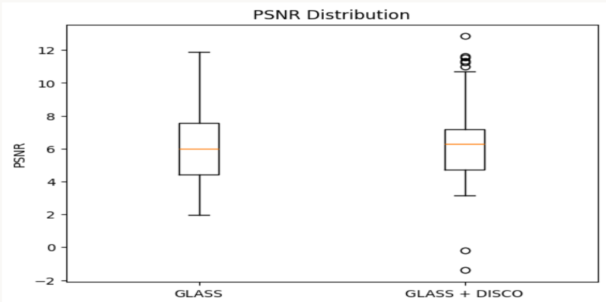
Quantitative Comparison





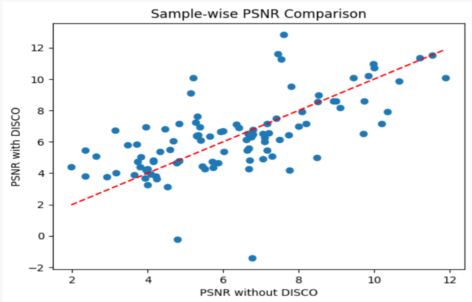
- Average PSNR with DISCO: ↑ 6.4 dB
- Average LPIPS with DISCO: ↓ 0.49

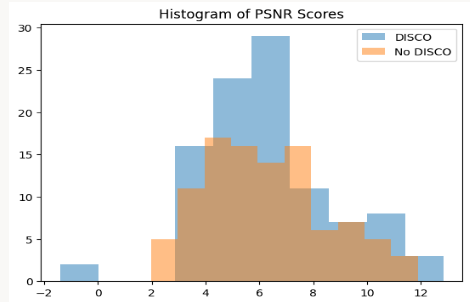
Sample-wise Analysis



- Boxplot and histogram confirm tighter and higher PSNR distribution with DISCO.

Distribution Analysis





- Most samples improve with DISCO (above red line).
- Outliers indicate instability without DISCO.

Conclusion

- WGAN is a lightweight substitute for heavy generators in GLASS.
- DISCO regularization enhances reconstruction quality across metrics.
- Future work: Extend to real-world surveillance and medical images.

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